CS4100
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Cracking the Code: AI for NYTimes Pips

A Comparative Study of CSP and Local Search Algorithms

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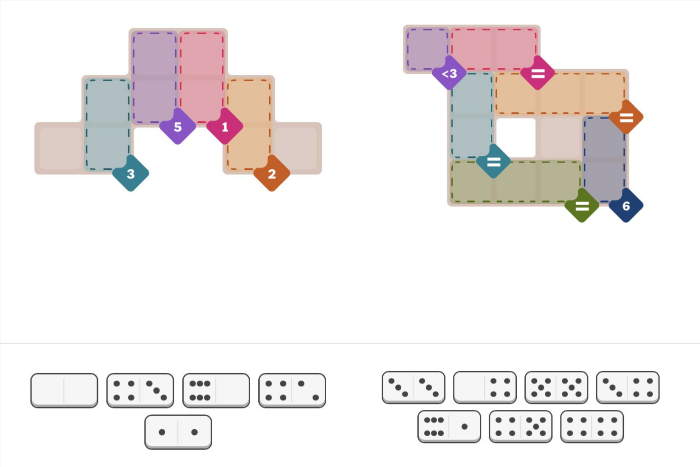
01

Introduction

What is NYT Pips? What is our project?

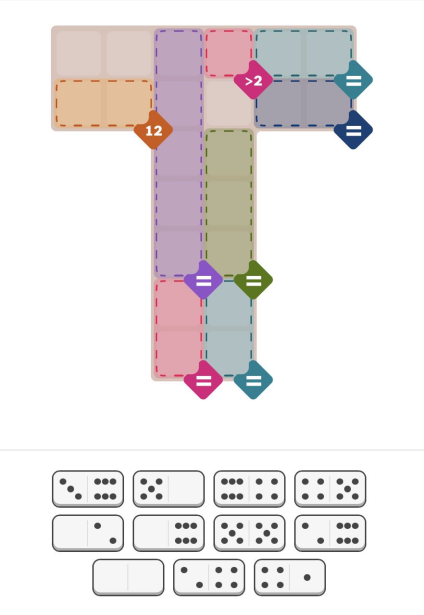
What is NYTimes Pips?

- New solo domino puzzle released by New York Times Games in August of 2025
- Players place dominoes on a grid under region-based constraints:
    - Sum
    - Equality
    - Inequality
- Difficulty scales from Easy (4–6 tiles) to Hard (14–16 tiles)



What is our Project?

- Build an optimal AI agent with the ability to solve all 3 difficulty levels of Pips
- Explore 2 AI strategies:
    - Local Search
    - Constraint Satisfaction Problem
- Evaluate which method best handles the constraint based grid game



02

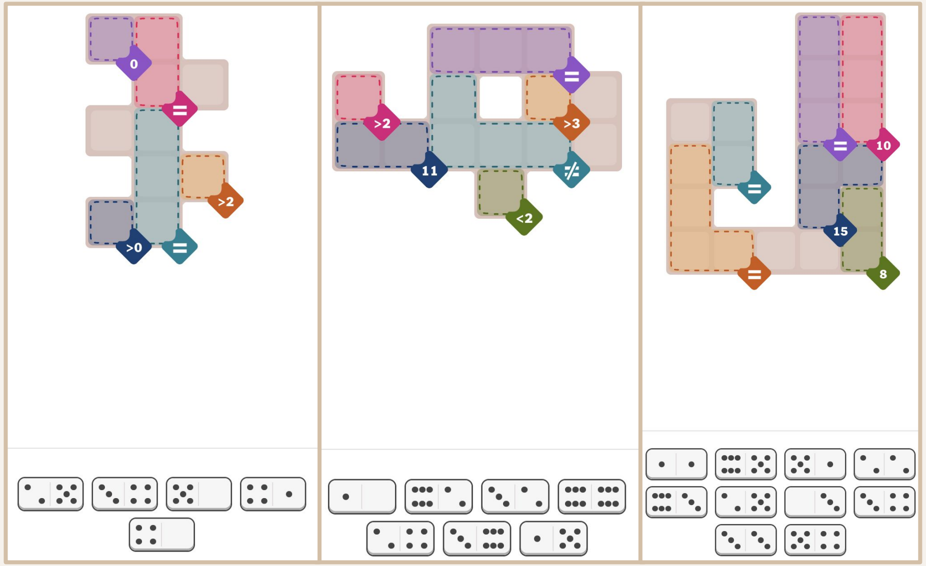
Strategy

How did we build this?

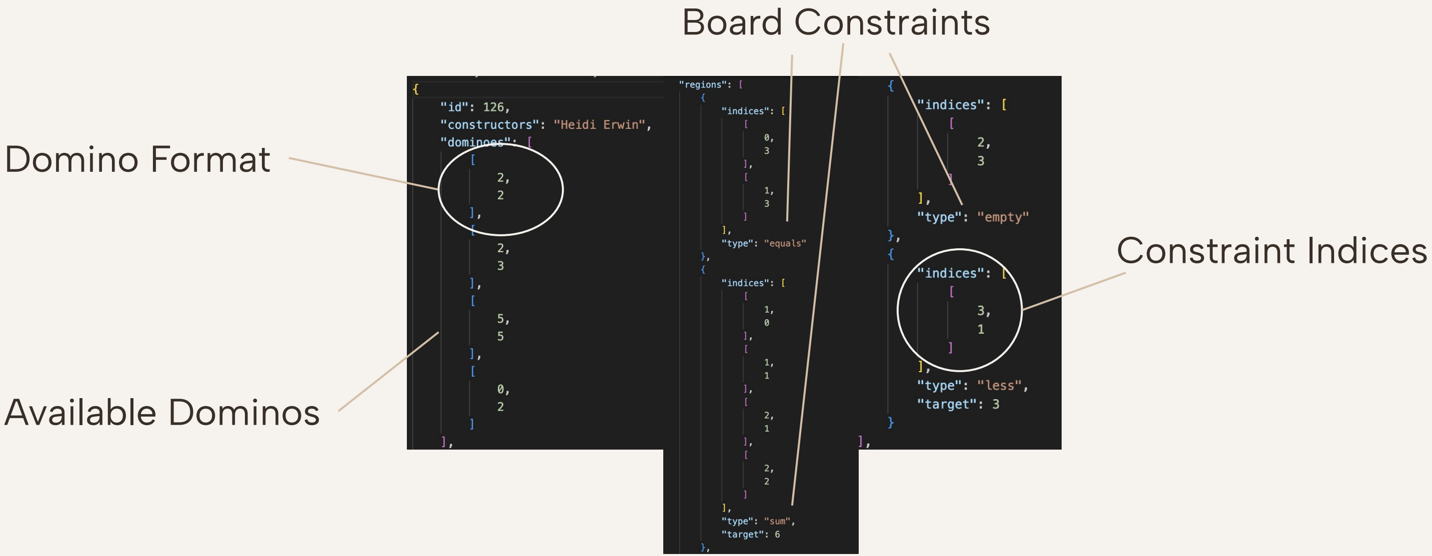
Building the Project

- Created mock versions of Pips for testing & solving
- Converted 200+ past Pips games into JSON objects
- Solved with two AI agents
- Created GUI for visualization

Easy Medium Hard



JSON Objects Explained



Approach & Algorithms Used

Local Search

- Random placement
- Simulated annealing
- Moves: swap, flip, re-tile
- Temperature-based probability
- Cost function for constraint violations

CSP

- Dominoes placed one at a time with backtracking
- Forward checking for remaining constraints
- Pruning strategies

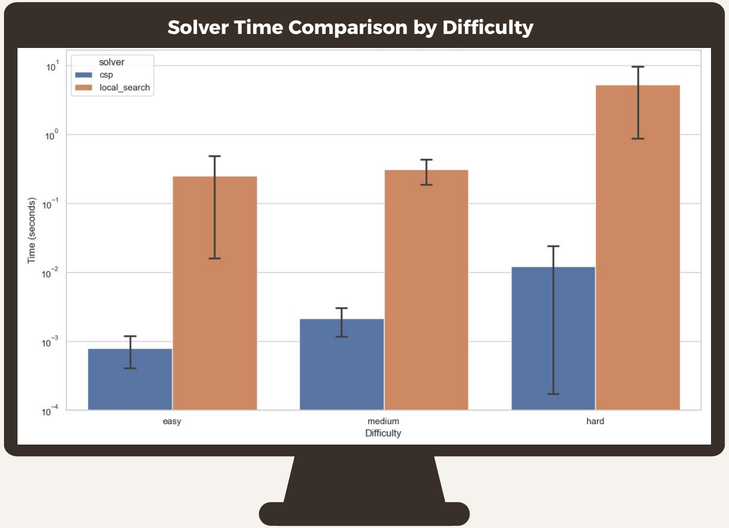
03

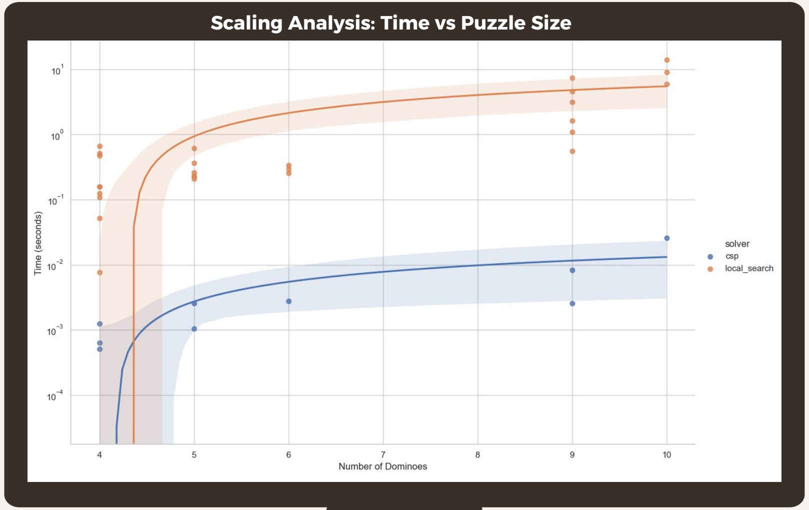
Results

Empirical Performance Analysis: CSP vs Local Search

CSP Dominates at Every Difficulty Level

- Easy: 1ms vs 250ms (250x faster)
- Medium: 1ms vs 250ms (250x faster)
- Hard: 20ms vs 5–10s (400x faster)



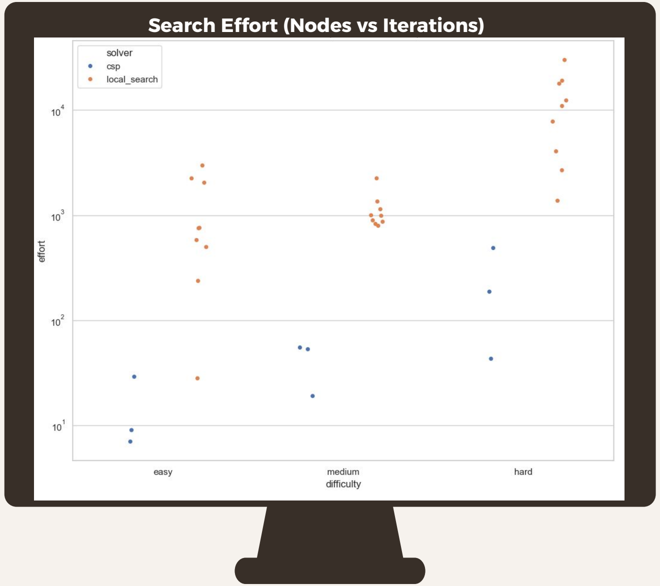


Scaling Reveals Exponential Crisis

- CSP: Linear growth (2x puzzle = 2x time)
- Local Search: Exponential explosion
- 15 dominoes: <1 second vs >10 minutes

Search Efficiency: 100x Difference

- CSP: 10–500 nodes with backtracking
- Local Search: 1,000–30,000 iterations
- Surgical precision vs trial-and-error



Whoa! Hypothesis overturned!

Expected: Local Search excels on constrained problems
Reality: Interconnected constraints favor systematic search

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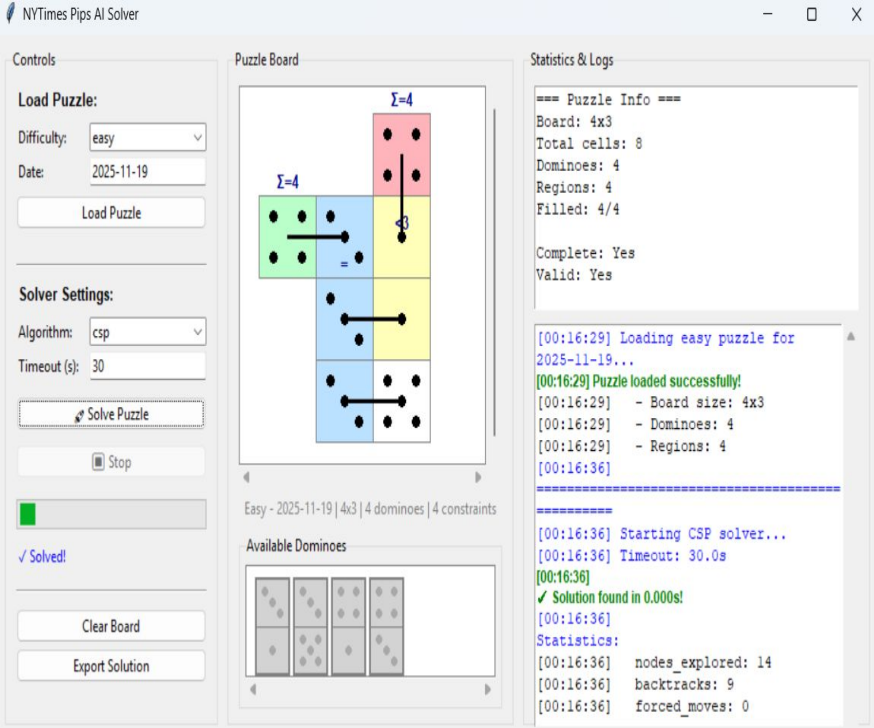
Project Demo

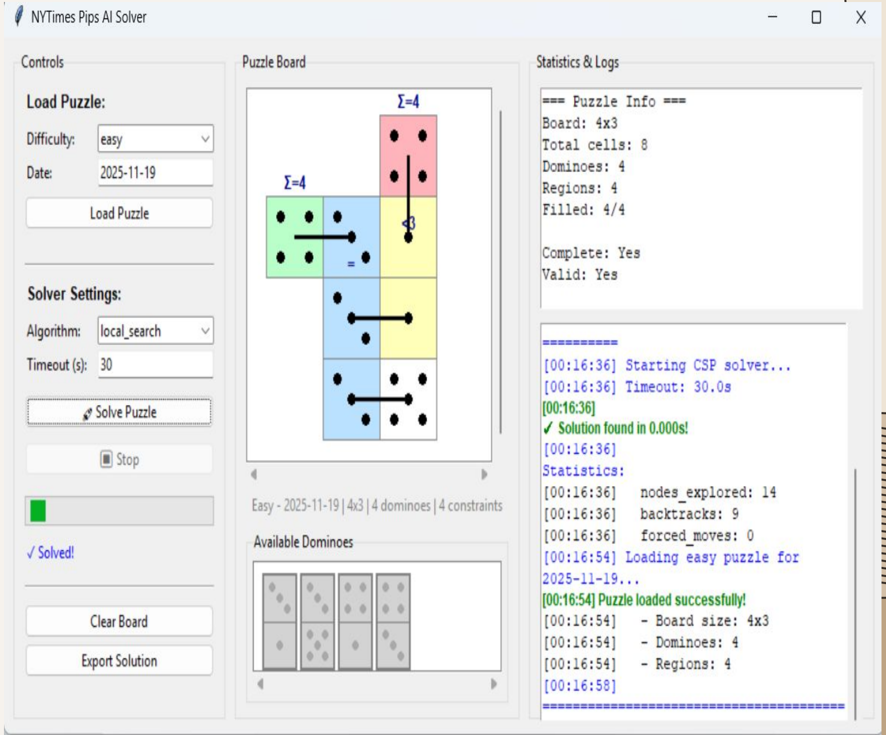
Visualizing Algorithm Performance Across Difficulty Levels

Live Demonstration

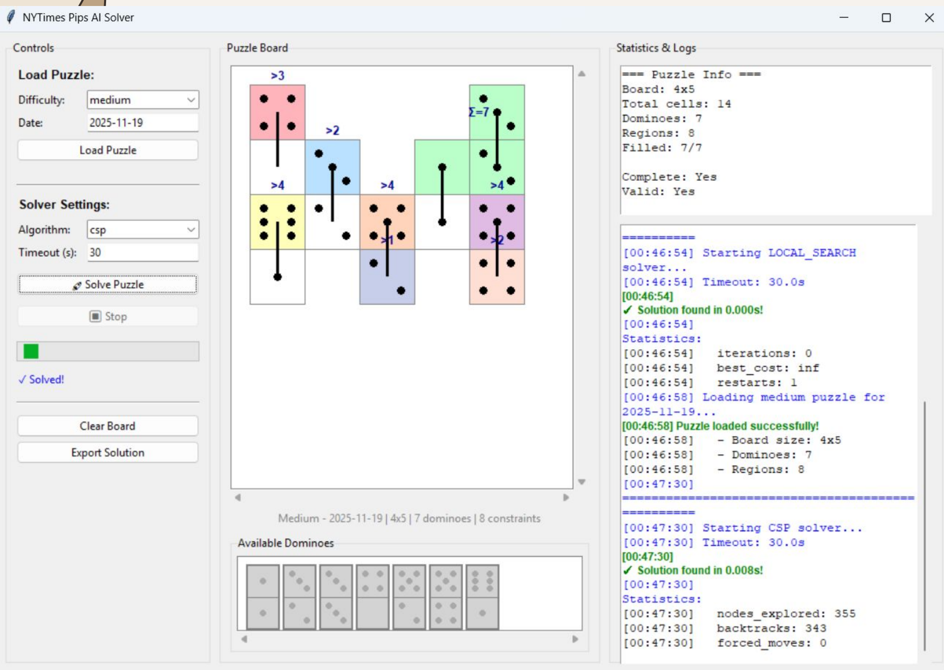
https://youtu.be/ME8ZCReth0w

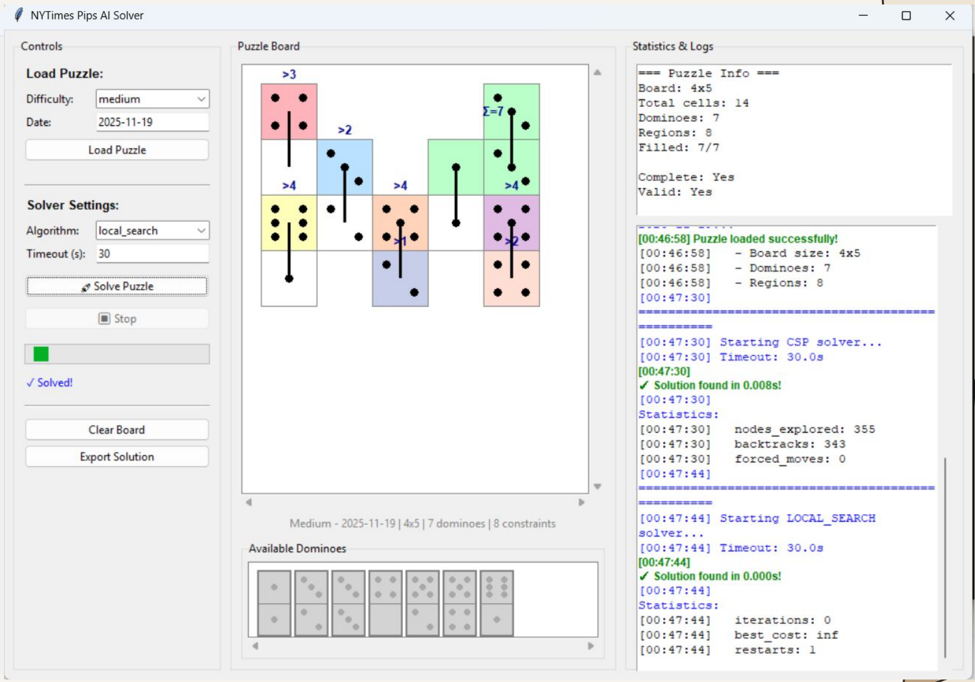
Easy Puzzle Performance



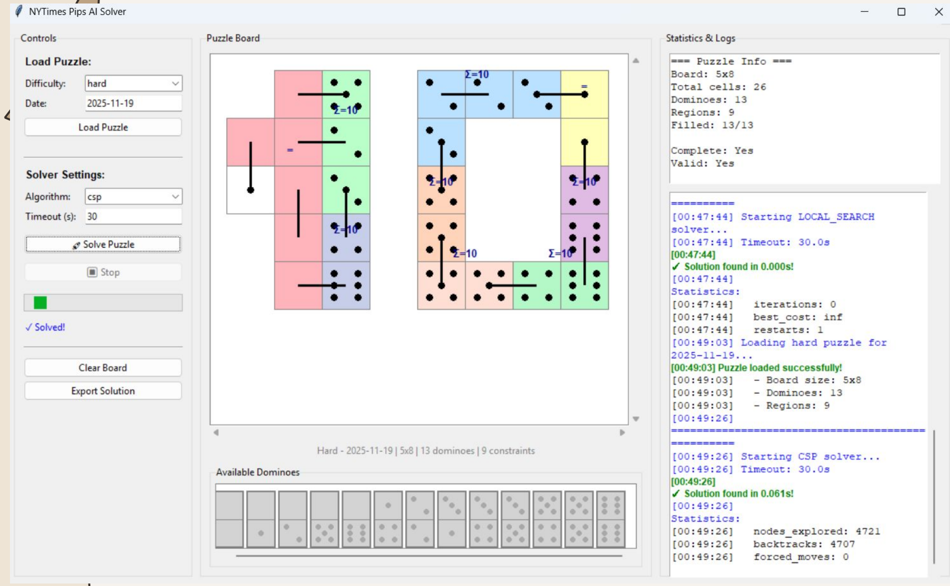


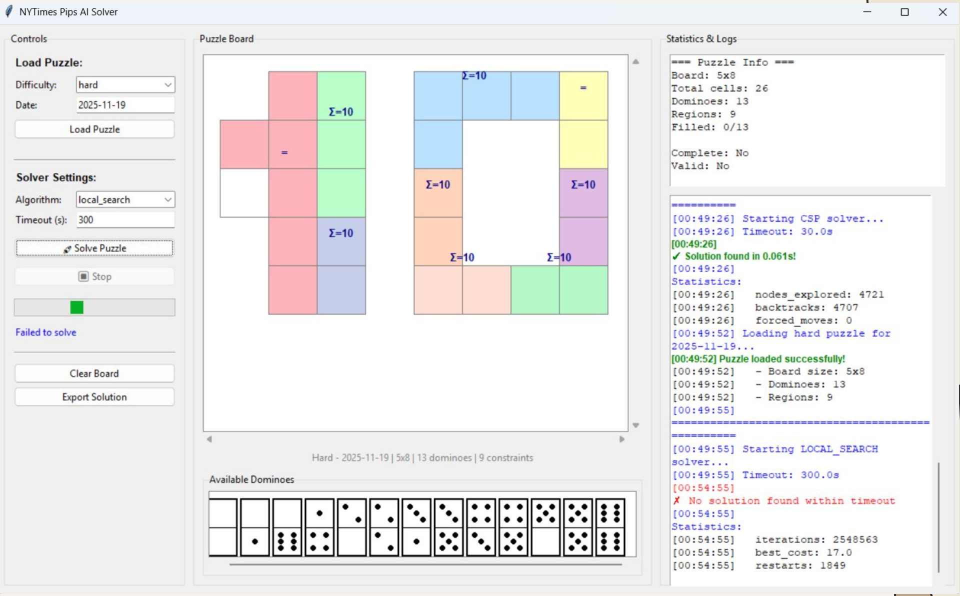
Medium Puzzle Performance





Hard Puzzle: The Decisive Gap





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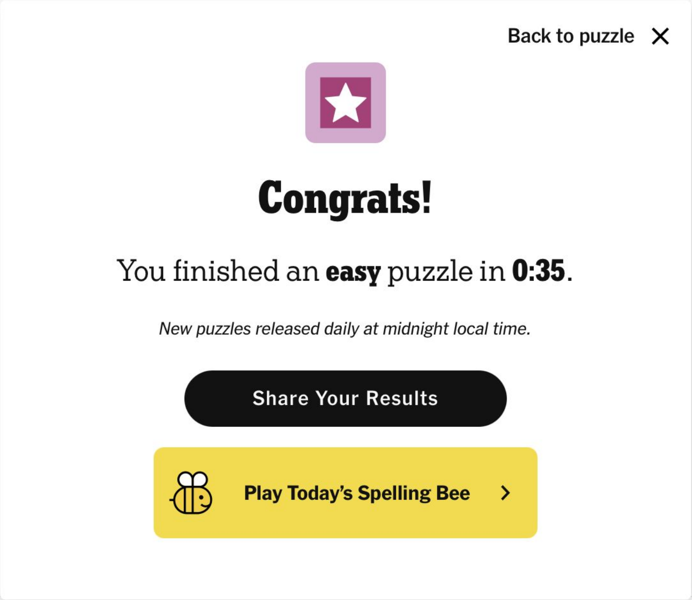
Conclusion

How can we move forward?

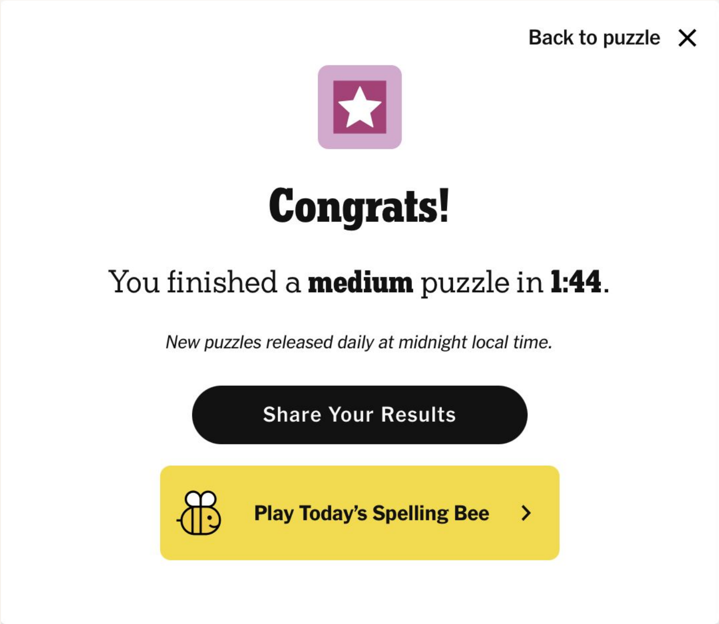
Key Takeaways

- CSP is the more optimal solver for NYT Pips
- Overlapping regional constraints align naturally with systemic search
- Local Search struggles because Pips penalizes partial progress
- Puzzle structure (not just difficulty) determines algorithm performance



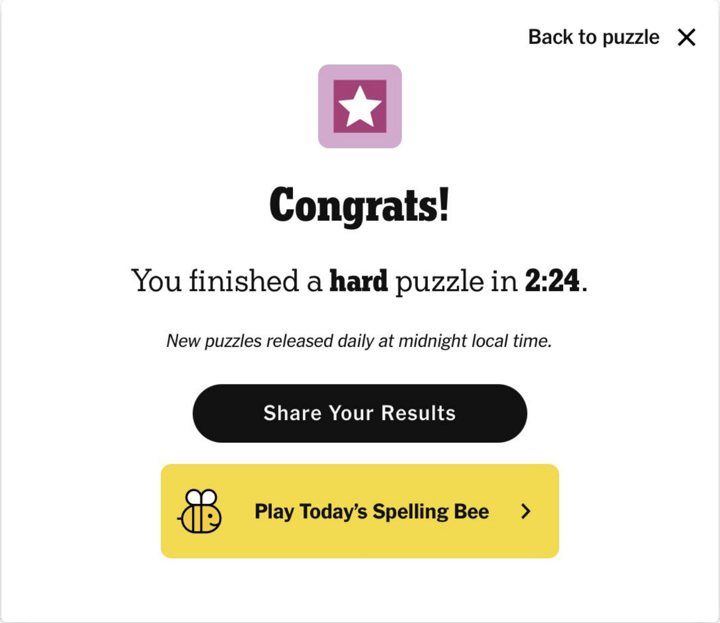
What we learned about Pips?

- Pips behaves like a true constraint satisfaction problem, not a noisy optimization task
- Constraint propagation yields major gains (especially on Hard puzzles)
- Performance scales gracefully with puzzle size when constraints are leveraged
- “Guess and improve” strategies fail when all regions interact tightly



Broader Implications

- CSP-based solvers could generalize to scheduling, logistics and layout tasks
- Local Search still holds potential with hybrid or informed heuristics
- Pips is a useful benchmark puzzle for teaching multi-constraint AI



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Questions/Comments

Thank you for listening to our presentation